UNIVERSITI TEKNOLOGI MARA

GAMIFICATION THEORY IN MOOC

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CHAPTER 1

INTRODUCTION

This chapter explains the detail and background of the study. Besides, problems of the subject under study are discussed and objectives are drawn based on the problems. Moreover, the details and significance of this study are listed.

1.1 BACKGROUND OF STUDY

MOOC stands for Massive Open Online Courses. MOOC is the free online courses available for anyone to use. Many universities use this platform to deliver their notes. MOOC is the online platform that designed for the large of number of students or participants to access their course. as long as the user has an internet connection, MOOC can be accessed by anyone and anywhere. It was to everyone without any qualification and fully for free to user use it (Jansen & Schuwer, 2015). MOOC is an online learning platform offering massive open online courses, which has become the world's largest online learning platform. The format to deliver the content lectures, reading a slide, and videos. There many examples of Massive Open Learning Course's (MOOC) which are Treehouse, Lynda, Duolingo, Codecademy, Education First, and Google Primer(Steve,2015). Massive Open Learning Course's (MOOC) are online courses that have the largest participants on the web.

Moreover, three basic types of MOOC learners are passive learner, bystander, and active learner. Thus, students can learn from anywhere and anytime at their comfort time to suit their learning style. If they don't understand, they can always repeat by going through the note, lecturer, or slide in the online platform as many times as they like. According to (Manisha,2020) there have many advantages of MOOC. Firstly, courses or content are offered in the MOOC for free to the student by the lecturer to use it. Secondly is, courses are offered in the MOOC are available for people from all over the globe. Thirdly, the performance of students that used MOOCs can be monitored easily by their lecturer using the

data captured during the start of courses. Fourthly, the lecturer and student can get world-wide exposure, thus it will improve pedagogical technique and knowledge in sharing. Lastly, the MOOC can be used as a tool in the blended learning program, where the student can access more content than what they get in the class.

In terms of learning in general, gamification theory is a theory of gamified learning. Gamification is defined as the use of the game attribute to increasing users' engagement during learning. In other words, gamification can be adopted to non-game contexts and activities by using game style mechanics(Investopedia,2019) As an example, Shopee has to make use of gamification theory in their marketing strategy. Without realizing it, we have played the game when we strive to get more rewards such as extra coins or vouchers. Similarly, there are many gamification used in learning such as create a quest, play digital games, and adapt old-school games for the classroom.

Gamification is different from gaming because gamification has good and useful content for the user. Gamification has the potential to increase students' motivation(Alsawaier, 2018). With the right design and implementation of the gamification method in learning materials, it could give a better learning experience, better learning environment, instant feedback, prompting behavioral change.

Based on the article, there are many advantages or benefits from the gamification theory in eLearning which can increase learner engagement. Gamification can give some opportunity for eLearning professionals to create some experience that engages their learners. The gamification can make their attention to that thing and can motivate them to reach their goal. The user would feel happy and positive when they were given rewarded for their effort. Thus, it able to make the user got long-term memory. Secondly, gamification would make eLearning interactive and fun. When the user has the objectives and some goal in their study, gamification theory is the most suitable because it can make learning not only informative but enjoyable and fun and exciting. To attract more users, the system can add an interactive element that can make the user exciting and more fun while they were studying. (Pappas, 2014).

For this project, the subject would use is Data Representation and Reporting Techniques (DCS119). This subject is all about data and visualization. Currently, the subject is not yet categorized as MOOC.

1.2 PROBLEM STATEMENT

In Universiti Teknologi MARA, MOOC was designed for students and lecturers. There are many context and content have been uploaded by lecturers to their students. The content in UiTM MOOC mostly is in the video for lecturing mechanism and text format for note as references. For video, most are designed with a voice as background to explain contented depicted in the slide presentation. This kind of method is good however, there is still room for improvement. Especially in the area to increase students' engagement during learning through the MOOC system. Besides, from our preliminary survey of FSKM students, they said that it was difficult to focus longer in their study because there are many slides in MOOC and make them easily bored. Moreover, an interview has been conducted with Dr. Fakhrul Hazman Yusoff, a multimedia lecturer from FSKM and year of service are for 22 years agree that the most of UiTM MOOC are lacking in interactive to UiTM MOOC for better engagement. Other than that, an interview with Dr. Masurah from FSKM UiTM Tapah. She is one of the MOOC evaluated panel, based on the previous MOOC evaluation process, she founds that some MOOCs are static and lack multimedia elements which unable to impress the reader. So I am believed this issue could be improved.

Next, the MOOC platform experienced a high rate of decline causing researchers to feel anxious from various aspects. they assume that MOOCs are less interactive which causes such incidents to occur. thus, they want to enhance the interaction within the MOOC platform to meet the needs of users and cause them to be more interested in using the MOOC platform (Gamage et al., 2020). Moreover, MOOC provides all the slides and video lectures along with the related reading resources. This method would make the student's scope for not

going through the lecturers only which does not lead to a deep understanding of the concepts(Michael,2014).

For the Data Representation and Reporting Techniques (DCS119), it is currently taught with lecture uploading the slide in UFuture (UiTM online learning system). Then lectures will either meet a student on line and explain the course or prerecorded the lecture. However, both means are without many multimedia elements. Moreover, the course is not yet categorized as MOOC.

Thus, this project aims to develop a MOOC for DSC119 with gamification theory embedded into it. This is to ensure high engagement by participants in learning the subject through MOOC.

1.3 OBJECTIVE

To develop the MOOC for DSC119, three objectives must be achieved which are:

1. To analyze various gamification theory
2. to develop a MOOC learning system for DSC119 with gamification
3. To test the user engagement with usability testing

1.4 PROJECT SCOPE

- This project will be focusing on the DSC119 course which is one of Data Science's courses.

- The focus is on three chapters in the course which is chapter 1-3
- The courseware was for the eLearning environment.

1.5 SIGNIFICANCE

The result from this project will enhance student's learning process through gamification theory. Gamification theory would make the student not boring to study using MOOC UiTM. The student would enjoy studying and the result is the student would understand better and their grade could increase. It also gives freedom to the student or lecturer to use MOOC UiTM since the system was an online platform, they can access it anytime and anywhere as long as they have an internet connection. This kind of system gives an opportunity to experience the course in gamification theory which attract student to use MOOC UiTM frequently.

CHAPTER 2

LITERATURE REVIEW

In this chapter, an overview of the important elements from past studies will be shown and briefly described. Most of these studies have been selected from a few resources such as journals, articles, and other resources that are related to this project development. This chapter will cover topics on education, courseware, the UiTM education system, and gamification theory.

2.1 LEARNING

Learning is the process or experience through from the experience or the problem-solving ability. The ability to reach the goal can make the learning increase through the experience (Washburne, 1936). Furthermore, For example, the student can understand and remember what they have learned and taught in class or online platform. In another word, learning is the transformative process is taking from the mixed experience than can changes what we know to builds something new. (Marcia,2010). There have four types of learning which visual learners, auditory learners, reading or writing learners, and kinesthetic learners. Visual learners for the student was easier to understand when presented to them in a graphic depiction of meaningful symbols. This student would make the mind map when they make a brief note. Auditory learners are more understanding when they are given the opportunity to hear information presented to them vocally. Reading or writing learners is the student when they were reading and writing, this method can make them easier to remember and understanding in their study. Lastly is the kinesthetic learners, it is hands-on. It's learned from the physically active role in the learning process. (Brianna,2019)

Nowadays, we can learn either face to face class or in the online platform. Face to face learning is an instructional method where the course content and learning material are taught in person or group of student. The befits of this method is they can learn face to face with their lecturer and the student can ask the question directly to their lecturer at that time. Other than that,

the student can gain a greater understanding and real-world examples from the lecturer and other students. But the disadvantages are there no many ways they can learn which they only can learn via the video presentation, teaching from their lecturer and they can get the content from the note or slide. This way, it can make the student easily bored with the study. But, learn from the online platform which online learning course, the student can study wherever they want at any time and anywhere (Tania,2017). Other than that, it was cost less which no need to ride a car to go to the class and avoid any traffic jam or accidents on the road. Moreover, they can easily repeat the study anytime if they don't understand the first round of study. Besides, the content can deliver in many ways for example in video, slides, and games. It can attract students to engage with their study better and they not easily bored to study. Thus, for this project, MOOC UiTM would use as an online learning course.

2.2 ON-LINE LEARNING SYSTEM IN UNIVERSITI TEKNOLOGI MARA (UiTM)

Nowadays, there are many online learning systems that UiTM provided which are known as i-Learn, UFuture, and MOOC. I-learn is a learning management system that offers resources such as course creation and management tools, content management tools, communication and collaboration tools, appraisal tools, personal information management tools, and system management tools. The system has seen a huge number of modifications for four years from the aspects of interface, usability, availability, functionality, presence of communicative tools, and software compatibility. (Adora,2011). Next, UFuture is a medium for online distribution of learning material to everyone who wants to take a course without any limitation on attendance. Last but not least, the Massive Open Online Course (MOOC) is a platform for the user to learn or study over the internet. Mostly, the contents in the MOOC is multimedia contents from a variety of course. It also encourages interactivity and using the tools available, the users can make any discussion or engage in other discussions(Mokhsin,2019).

2.3 MASSIVE ONLINE OPEN COURSE (MOOC)

For this project, MOOCs would use as an online platform. this kind of online learning system is the online platform for the lecturer and students to communicate more about the learning. There a certain of UiTM students are use the MOOC to access notes given by their lecturers. The notes given in the MOOC are in the form of the slide and video elements. Thus, most of the faculty in UiTM was used this platform as blended learning as one of the learning methods. However, some students find it difficult to get engagement with the MOOC because students easier bored to learn because the content in the MOOC was in the slide or video element only. Also, based on an interview that has .doned with Dr. Masurah from FSKM UiTM Tapah. She is one of the MOOC evaluated panel, based on the previous MOOC evaluation process, she founds that some MOOCs are static and lack multimedia elements which unable to impress the reader. So she believed this issue could be improved. Other than that, an interview has been conducted with Dr. Fakhrul Hazman Yusoff, a multimedia lecturer from FSKM and the year of service are for 22 years agrees that most UiTM MOOCs are lacking in interaction to UiTM MOOC for better engagement. This is because the learning method is static and lacking interactive and multimedia elements. Therefore, there is a method that can use in more interactive learning and it is called gamification theory.

2.4 GAMIFICATION THEORY

Gamification theory describes the incentivization of people have an engagement with non-game contexts and activities by using game-style mechanics. Gamification leverages people's natural tendencies for competition, achievement, collaboration, and charity such as rewarding users for achievements, earning badges. This method can help to motivate individuals to achieve their goals or boost performance. The evidence of success that can be measured success is through the involvement, influence, time spent on the activity. (Investopedia,2019). Besides, gamification could be supported by various platforms, such as tablets, computers, and smartphones.

2.5 GAMIFICATION ELEMENT

Gamification was designed simply by using game design elements in non-game contexts. (Szeto, 1991) .There are many elements in gamification such as achievement(Progression), rewards, story, and time. There are many examples for each gamification element such as for achievement are points, badges, leveling, leaderboards, progression bars, and certificates (Mindy,2017).

2.6 GAMIFICATION TAXONOMY

In term of elements can be applied in an educational environment, there are few taxonomies was valid in the gamification theory. The elements of gamification taxonomy are performance, ecological, social, personal, and fictional. (Oliveira,2019).



Figure 2.1 Gamification Taxonomy

2.6.1 PERFORMANCE

These are elements related to the environment response which can be used to provide feedback to the user. In this element, there have points, progression, level, stats, and acknowledgment. The acknowledgment is also known as medals, achievements, and badges (Klock et al., 2018). Next, the level is also known as skill level, character level. When the user completes certain tasks, they would gain a level and would get more challenging tasks. Moreover, the progression is also known as steps, maps, and progress bars. It guides the users of their advance and allows these users to locate themselves. Besides, the point is also known as scores, skill points, and experience points. This element was the most basic concept in all gamified applications (Dichev & Dicheva, 2017). Lastly, stats are also known as information and data. It would show the overall stats in virtual environments such as dashboards (Koivisto & Hamari, 2019).

2.6.2 ECOLOGICAL

This context is related to the environment that the gamification is being implemented and can be represented as properties. The elements in this context are economy, chance, imposed choice, time pressure, and rarity. Chance is also known as luck, randomness, probability, or fortune. This concept is related to the random property of a certain event. For example, spinning roulette would give a special bonus based on their luck (Dignan,2011). Secondly, the imposed choice is also known as judgment, paths, and choice. This concept is when the user has to make any decision and needs some advice in a certain environment. Thirdly, the economy is also known as the market, transactions, and exchange. This concept is related to the situation that has any transaction in the environment. Next, rarity is also known as the collection, limited items, and exclusivity. It is related to the limited resources within the environment which the user has to through a specific goal. Lastly, time pressure is also known as clocks or countdown timers. It is related to time which used to give some pressure on the users to take extreme action to finish their tasks (Toda, Oliveira, et al., 2019).

2.6.3 SOCIAL

This dimension is related to the interactions between the user present in the environment. The elements in this dimension are competition, cooperation, reputation, and social pressure. Firstly, competition in other words is also known as leader boards, conflict, and scoreboards. This concept is when the user faces another user to achieve the same goal using scoreboards based on their levels, points, and badges. Secondly, cooperation is also known as groups and teamwork. The concept for these elements is when the user must collaborate with other users or make a group to complete the task given. An example of this element is where tasks have made the groups interact with each other and recognized by these interactions (Shi et al., 2014). Next, reputation is also known as status or classification. It is related to the titles that the user may gain within the environment. These titles are usually used within the users to create a hierarchy in the environment. Lastly, social pressure is also known as guild missions or peer pressure. This concept is related to the social interactions that give pressure on users.

2.6.4 PERSONAL

For the personal, this dimension is related to the user that is using the environment. The element in this dimension is sensation, objective, novelty, puzzle, and renovation. Firstly, the novelty is also known as the changes, an update, or a surprise. This element was related to the update that occurs within the environment by adding new content or information (Hanus & Fox, 2015). Secondly, the objective is also known as missions. This concept is related to the goals. It provides the purpose to the user to perform the task given (Toda, do Carmo, et al., 2019)(Alsawaier, 2018). Thirdly, the puzzle is also known as cognitive tasks, challenges, or actual puzzles. This concept is related to the activities that are implemented within the environment. This element was suitable for education through quizzes or challenges. Next, renovation is also known as extra life, renewal, or boosts. This element is related to the situation that the user can re-doing a task. It gives a chance to the user to make a better result in a certain task and would make that games would interesting and fun (Hammer,2011). Lastly, the sensation is either sound of simulation or visual. It is related to the user to improve their experience. But,

for this element, it can be done through gameful interfaces and dynamic such as augmented reality (AR) or virtual reality (VR).

2.6.5 FICTIONAL

Fictional is the mixed dimension that is related to the user through narrative and environment (through storytelling). There have two elements in this dimension which narrative and storytelling. Firstly, the narrative is also known as implicit decisions or karma system. this concept would give the user experience the same goes as happen in the game. This experience would be influenced by an implicit choice made by the user (Palomino et al,2019). Lastly, the storytelling can be as the text stories and audio queues. It is the way of the story of the environment, in other word is a script. It would through the text, sensorial resources, or voice (Palomino et al,2019).

2.7 COMPARISONS OF GAMIFICATION TAXONOMY

This section compares the five gamification taxonomy which has been discussed above. Their aims, weakness, and strengths are also highlighted in the section.

Table 2.1 shows the comparison between gamification taxonomy in their aim, strengths, and weakness

Taxonomy/ Features	**Performance**	**Ecological**	**Social**	**Personal**	**Fictional**
Aim	To provide feedback to the users.	Motivate the user to proceed with give some bonus	Interactions between the user presented in the	The users can train themselves to be more independent.	The user can create great scripts.

			environment and have a grouping task		
Strengths	The user can see their point, know the level.	The user can get some bonus from their games.	Can improve the attitude of cooperation because they have to play in groups	The user does not have to make a group or team complete the task and can make their own decision.	The user can learn more about writing.
Weakness	Some users will feel depressed if they cannot get the targeted point.	These elements make the environment feel dull, as it does not have elements that produce interactions with the user.	These elements make the users always have to create a group to complete the task.	This element makes the users will be isolated since they will not be able to interact with other students.	The user would lose the meaning of context within the immersive environment because the user has to perform any task because influence the quality of the user experience.

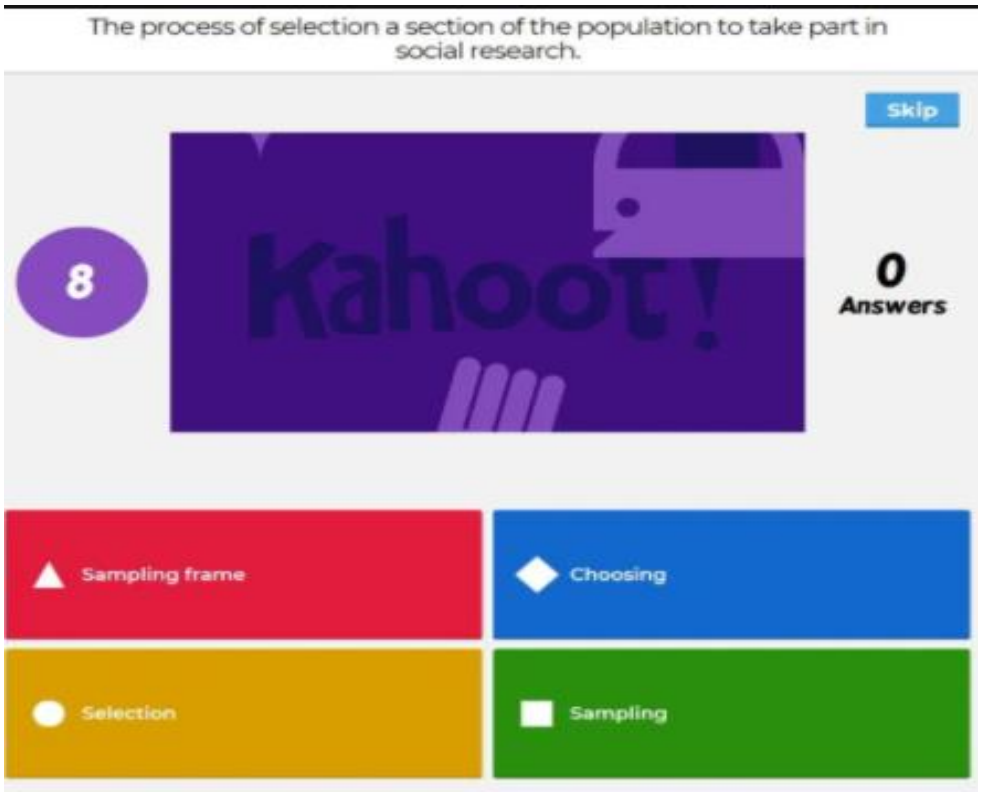
Based on all gamification taxonomy compared above, it seems that performance is more suitable to be used because the performance has the best element which has levels, point, progression, stats, and acknowledgment. For education, the elements in the performance suitable because it can help the user engage and have the motivation to study.

2.8 SIMILAR APPLICATIONS

Other similar applications used performance as an element in designing gamification theory in MOOC. Some of the systems are Kahoot!, Quizlet and Quizizz. Depicted below is the analysis of the systems.

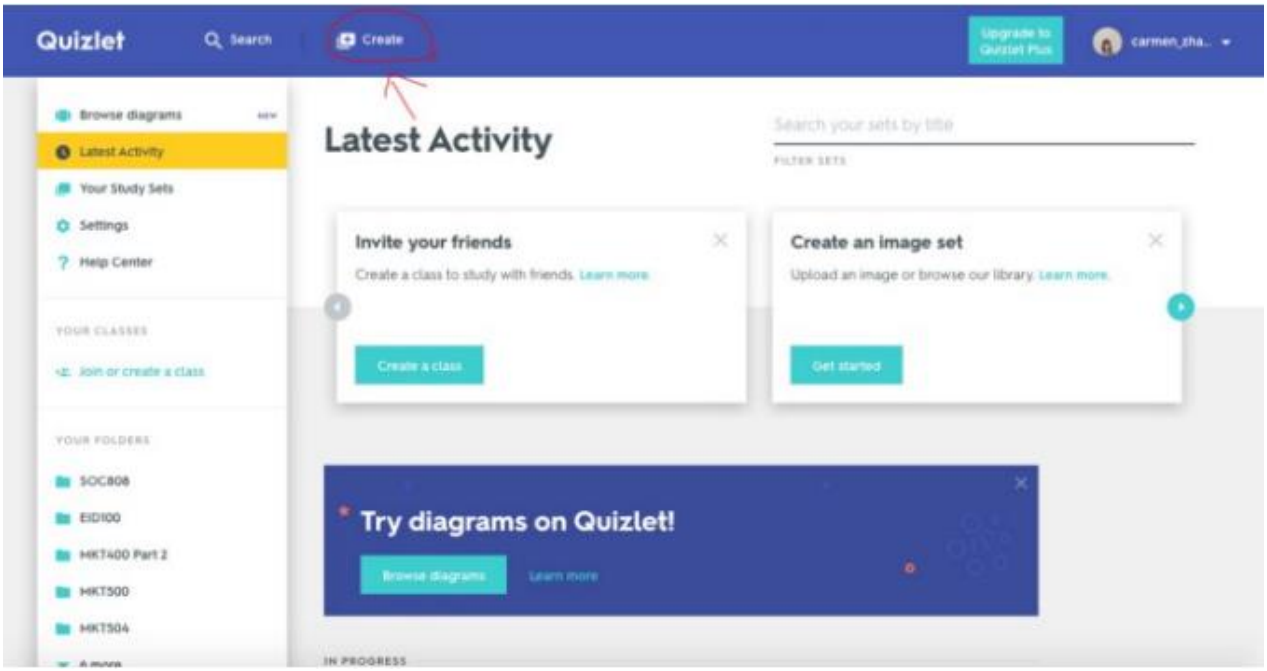
2.8.1 Kahoot!

Kahoot! is a game-based learning platform that easy to share, create, and play a learning game. It was used as educational technology in schools and other educational institutions. Kahoot! is a learning game were have multiple choice quizzes that the user can access via the Kahoot app or a web browser.



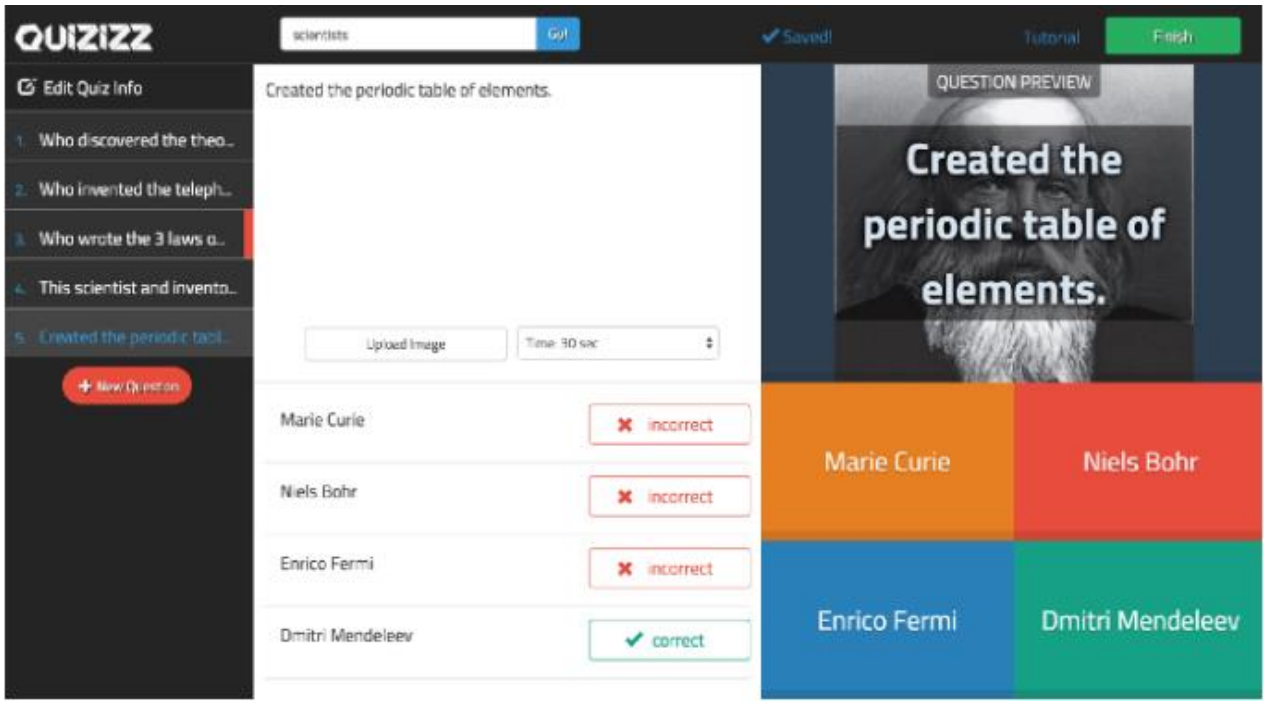
2.8.2 Quizlet

Quizlet is a free website is a free website providing students with learning resources, including flashcards, game modes, study modes. It was founded in 2005 by the high school sophomore Andrew Sutherland. With terms and definitions, you can start by creating your own study sets.



2.8.3 Quizizz

Quizizz would help the user to conduct student-paced formative assessments for students of all ages in a fun and engaging way. The question would appear on each user's screen and it would easier for the user to answer questions at their place and review their answer at the end of questions.



Based on Table 2.2, a comparison of each similar application shown below is based on what each system application has or does not have based on its elements.

Table 2.2 Comparison of what each system haves and do not have

Elements/ Features	**Kahoot!**	**Quizlet**	**Quizizz**	**Gamification theory in MOOC (DSC119)**
Levels	no	yes	no	yes
Progression	yes	yes	no	yes
Point	yes	yes	yes	yes
Acknowledgment	yes	no	yes	yes
Stats	yes	yes	yes	yes

In Table 2.2 show that the existing application is lacking some features. therefore, in gamification theory in MOOC (DSC119) system will be the features that the other application is lack.

2.9 SUMMARY

Gamification theory has been widely used and applied worldwide also in the education field. Gamification theory plays an important role in the education system to make the students more interested and not easily bored to learn and study.

For the elements, performance will be chosen as a gamification theory in MOOC because the element itself is more suitable for education learning. Other than that, easier for the lecturer to handle it and the student would more interesting to use the MOOC platform to study.

CHAPTER 3

RESEARCH METHODOLOGY

3.0 Introduction

This chapter describes the project methodology as the procedures used in developing the prototype. It consists of the project research framework and workflow that are submitted to the development phase.

3.1 Project Research Framework

In this chapter, the detail about the methodology that is going to be used in the project is discussed. It explains how the method works during the development process. The waterfall model is one of the courseware development models that was chosen as the instructional model. Since, the project focus on the gamification method and learning, thus the Waterfall model is the most suitable model instructional model to complete this project. In the Waterfall model, there have five phases which are requirements, design, implementation, verification, and maintenance.

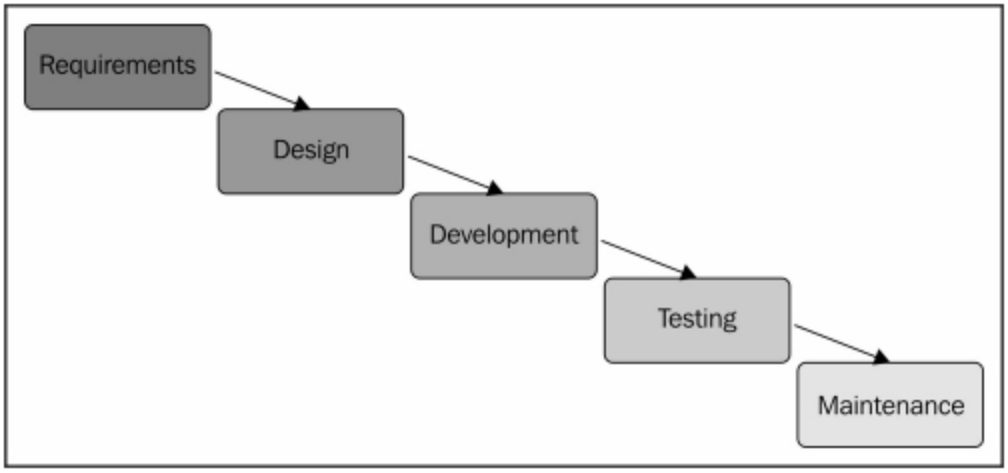


Figure 3.1 Waterfall Model for Game Development

3.2 Requirements

Based on the research and observation the problem statement of this project as stated in chapter one have been identified. The solution to this problem and objectives of this project also has been determined to develop a more enhanced for the gamification theory in MOOC. The main problem is the MOOC lack of interactive multimedia and engagement with the system which only have such as a slide and video conduct by the lecturer. Thus, gamification theory would be used to make it more interesting and make the user would engage more in the system. By going through the article related to the gamification theory, a different type of gamification taxonomy has been discovered. By discovering all these taxonomies, it can help in developing this project. MOOC will be more engaged with the system and more interactive for the user to use the system.

Next, the information for the project was collected from the collection of journals, books, and articles that are related to the gamification theory in e-learning, the various technologies used which gamification taxonomy, and other resources that are related to the project. The gamification taxonomy is performance, ecological, social, personal, and fictional. Thus, only one gamification taxonomy has been chosen and the most important process is to discover a performance that is suitable for the project requirement.

3.3 Design

3.3.1 Use Case Diagram

In the design phase, the design of the overall project will be developed on the overall use case diagram shown in **Figure 3.2**

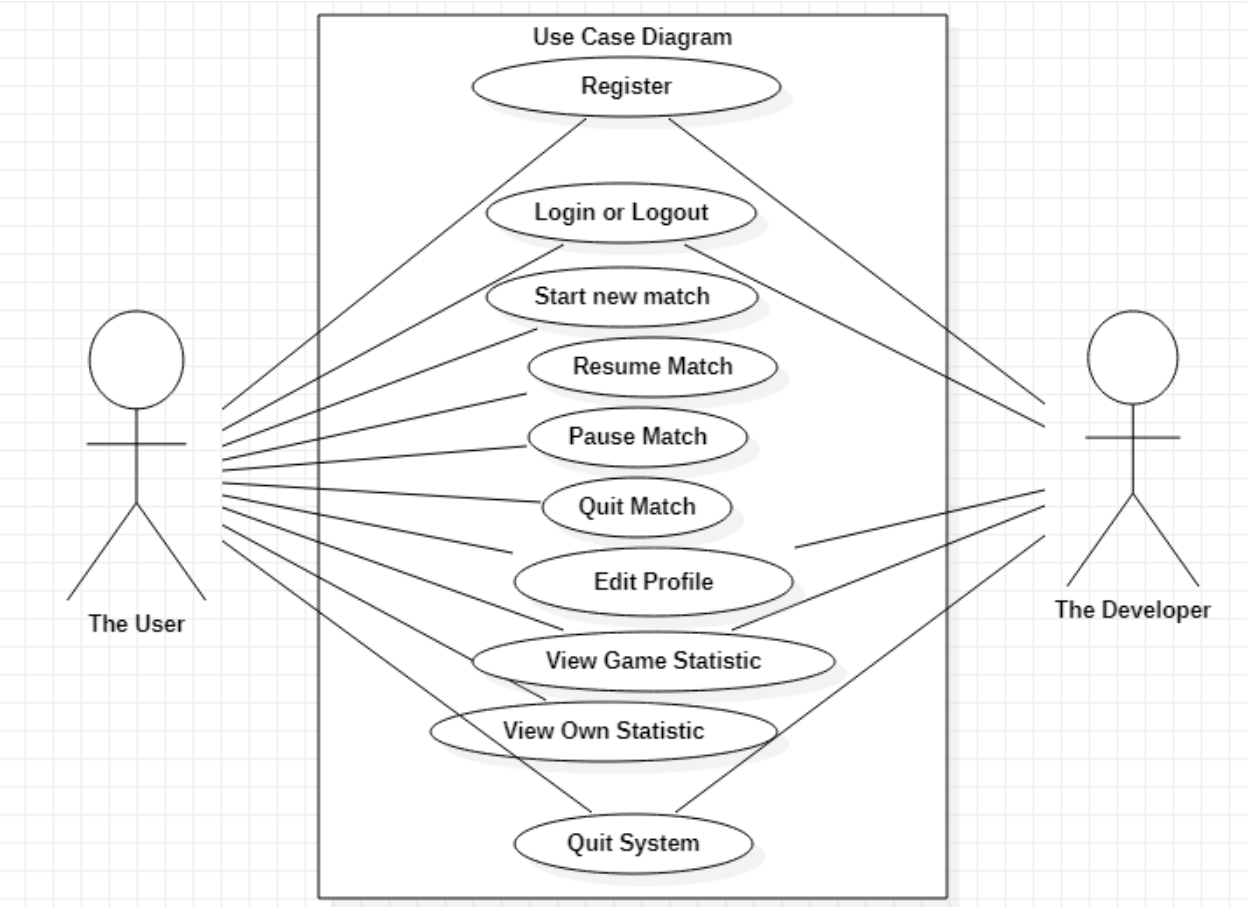


Figure 3.2 Use Case Diagram

There have two actors which are the user and the developer. There are nine use cases in the diagram which are register, login/logout, start a new match, resume the match, pause match, edit profile, view game statistic, view own statistic and quit. The user can explore all the use cases but for the developer he or she can only five use cases only which are the register, login/logout, edit profile view game statistic, and quit the system. When the user running the system, the system will display the login page which the user has to insert their username and password to

enter the system. after success enters the system, the system would display the main page with two-button which start a new game and resume the previous game. When the user plays the game, the user can pause the match, quit the match, and resume the match when the user paused the game. Other than that, the user can edit their profile such as their email and new password. Also, the user can see their own score or overall score for the game. Lastly, the user quit the game with the quit system.

3.3.2 Overall Flowchart

In these phases, this project required a flowchart to show the flow of this project. The flowchart will determine the look, the design, and the content of the project.

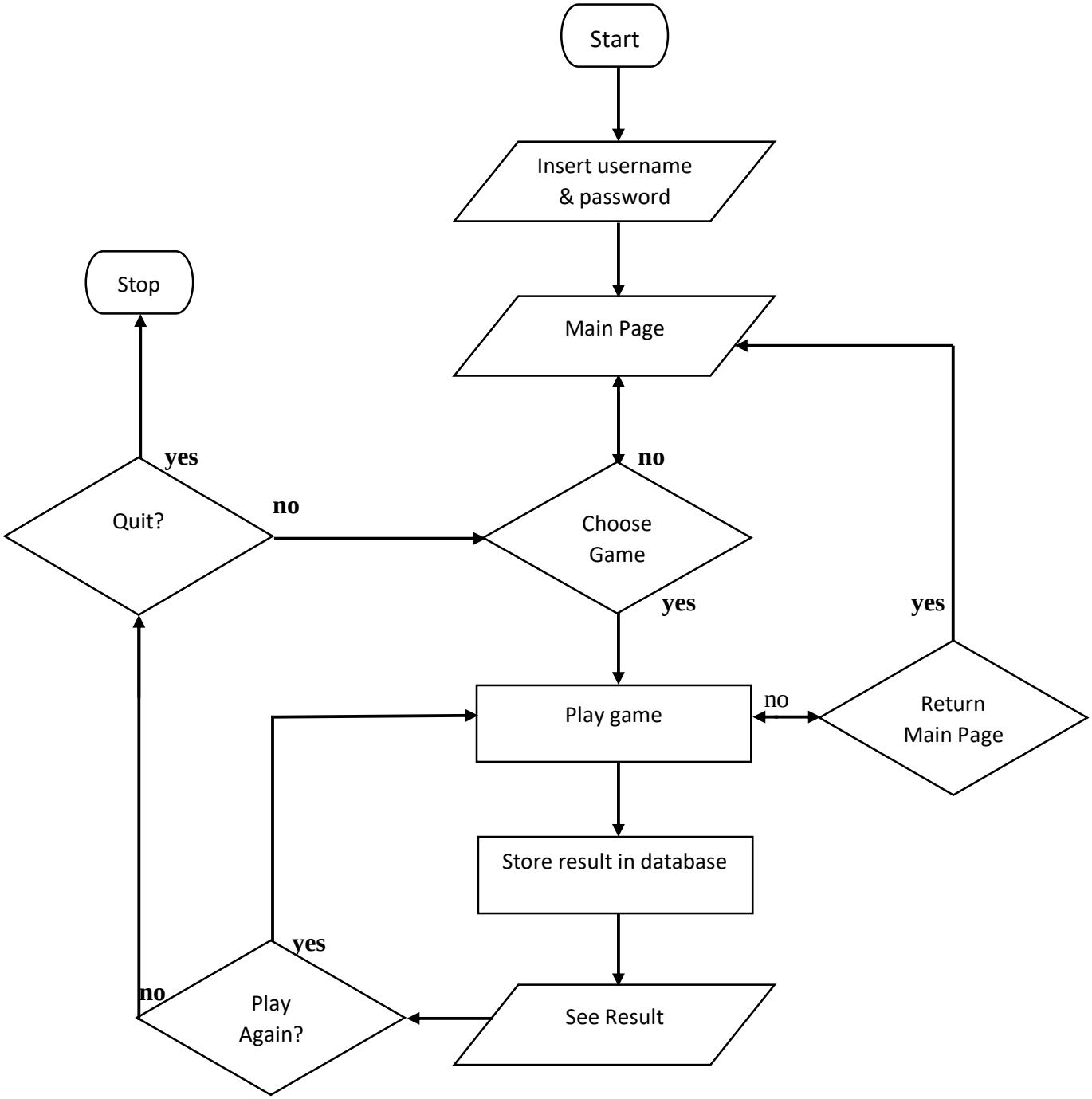


Figure 3.3 Overall Flowchart

Based on Figure 3.3 above, the main flow chart contains the start/end, the input/output, the process, and the decision. There are 2 processes, 3 input/output, and 4 decisions that the user is going to choose. In this flowchart, firstly is the input which the user has to insert their username and password at the login page. Next is the output in which the Main Page would appear. After that is the first decision which the user has to choose either want to play a game or not. If the user wants to play, the first process which plays game process would be applied but if the user does not want to play the game, it would static at the Main Page. While playing the game, there has the next decision which if the user wants to return to the Main Page. If the user chooses yes, the Main Page would appear but if the user chooses no so the user would continue as usual to play the game. After the user finished the game, the result which the score would be stored in the database. Next, the first output which the result from the play would appear. Furthermore, the third decision is the user wants to play again the game or not. If the user chooses to play a game, it would go to the play game process but if the user chooses no so the last decision would appear which if the user wants to quit the game or not. If the user chooses yes so automatically the user would log out from the system. but if the user chose no so it would go to the Main Page.

3.3.3 Flowchart for Login and Register Flow

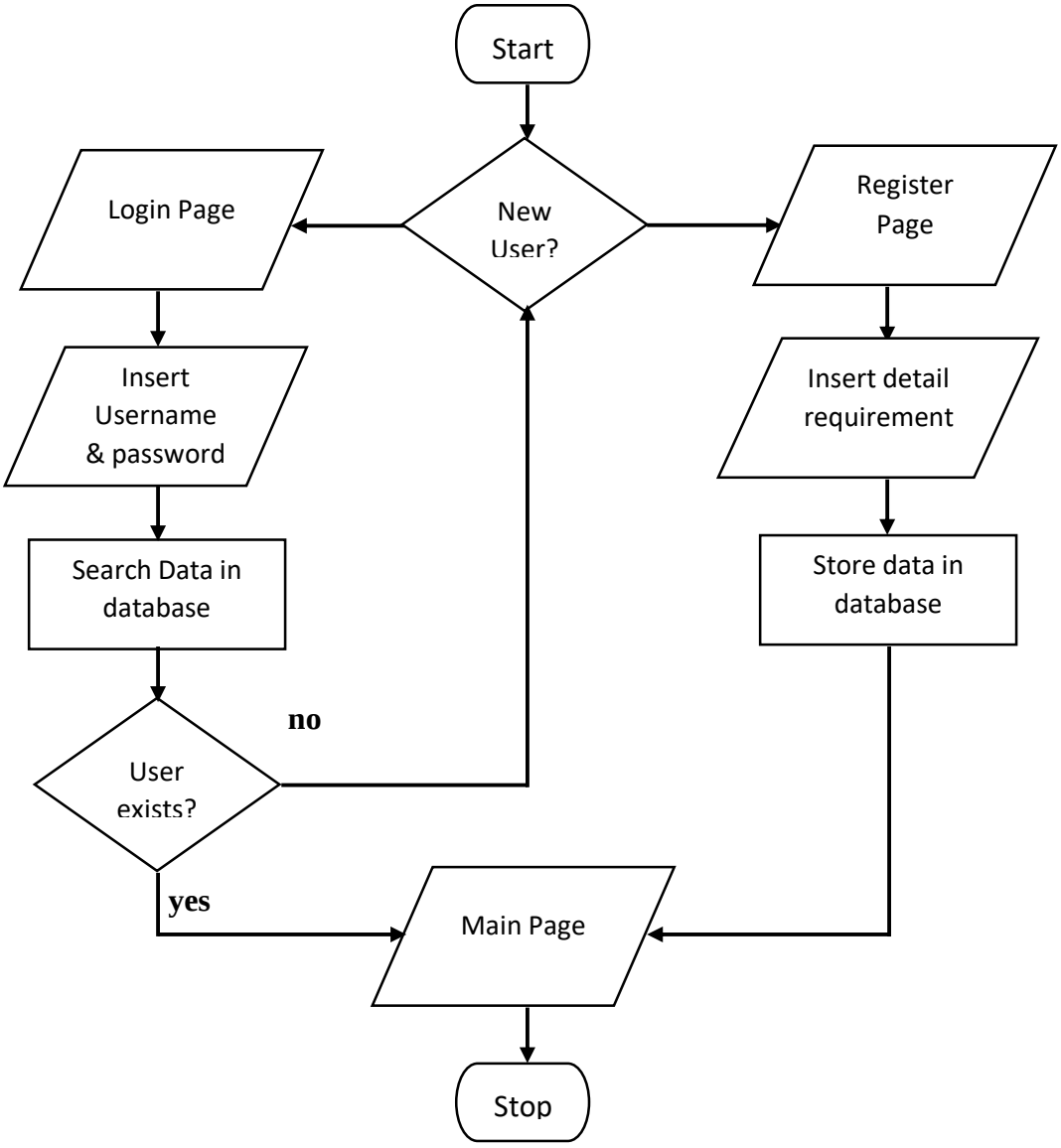


Figure 3.4 Flowchart for Login and Register Flow

Figure 3.4 shows the process flow diagram of a login and register. It contains the start/end, the input/output, the process, and the decision. There are 2 processes, 5 input/output, and 2 decisions that the user is going to choose. For this flowchart, firstly is the first decision which if the user is the new user, he or she has to register but if not he or she has to log in. For the login, the user has to insert their username and password to enter the system. The system would search their data in the database. If their data existed, so automatically they would enter the system to the main page

but if not, they have to insert again their username and password go through with the same process. Next, for the register, the user has to fill in the requirements which their name, gender, age, email, username, password, re-enter a password, and ID student. After that, the system would store the data in the database. Then, automatically the user can enter the system.

3.3.4 Flowchart for Game

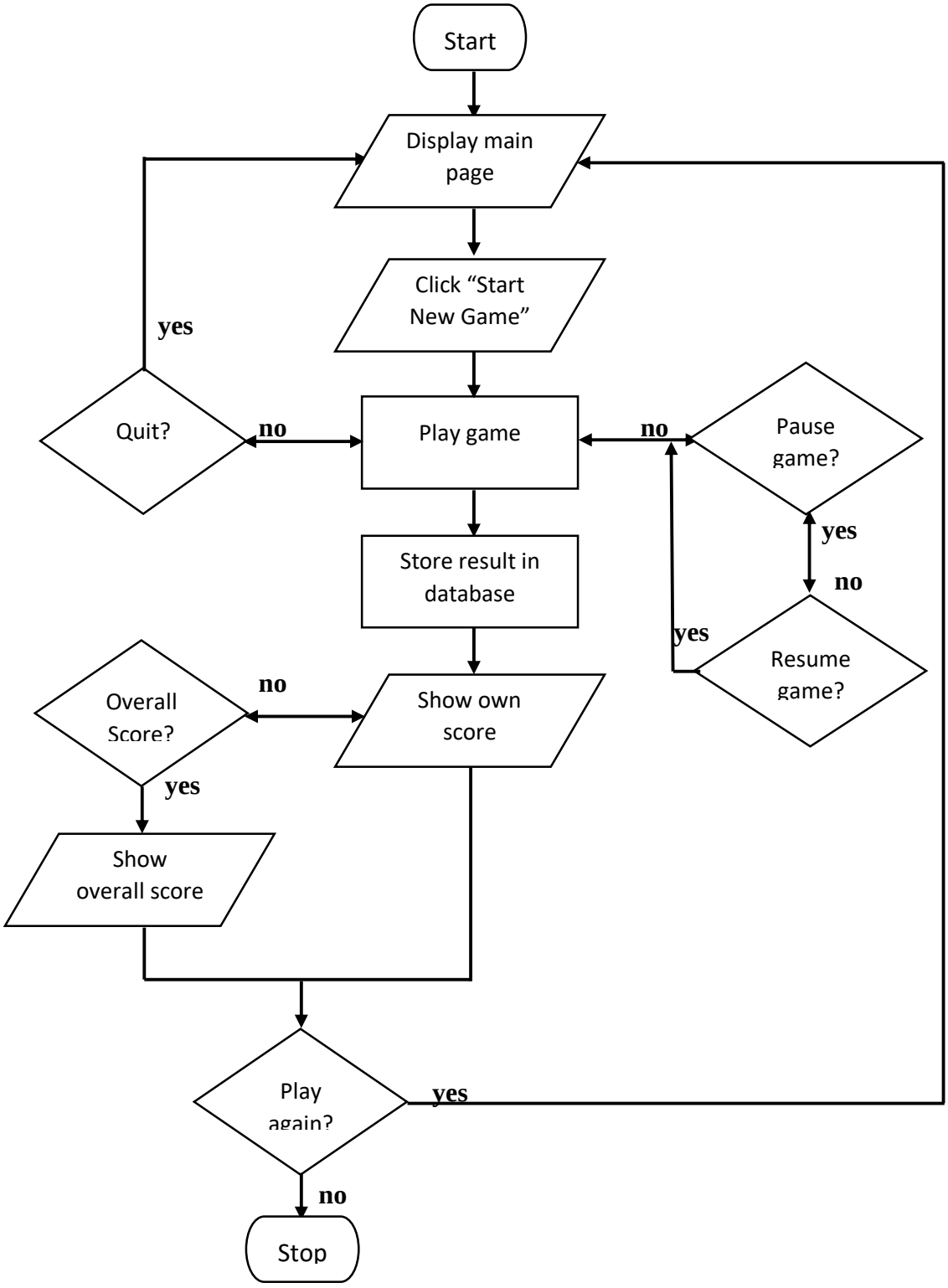


Figure 3.5 Flowchart for Game flow

Based on Figure 3.5 is the flowchart for game flow. It contains the start/end, the input/output, the process, and the decision. There are 2 processes, 4 input/output, and 5 decisions that the user is going to choose. Firstly, on the main page, the user has to click on the “Start New Game” button to start the game. While playing a game in the play game process, there have 3 decisions which quit, pause, and resume. For the quit, if the user wants to quit from the game, it would go to the main page but if not it will stay t the play game process. Next is for the pause decision. If false, so it would stay at the play-game process but if not it would have another decision that resumes decision. If false, the game would pause but if true the user can continue playing the game. After finished play the game, the score would store in the database, and output for the own score would appear. in this output, there has one decision that shows the overall score. If true it would show to the user the overall score that has been collected by the database from other users. Furthermore, there one decision that asks want to play again or not. If the user chooses to play again, it would go to the main page but if not it would quit the system.

3.3.5 Class Diagram

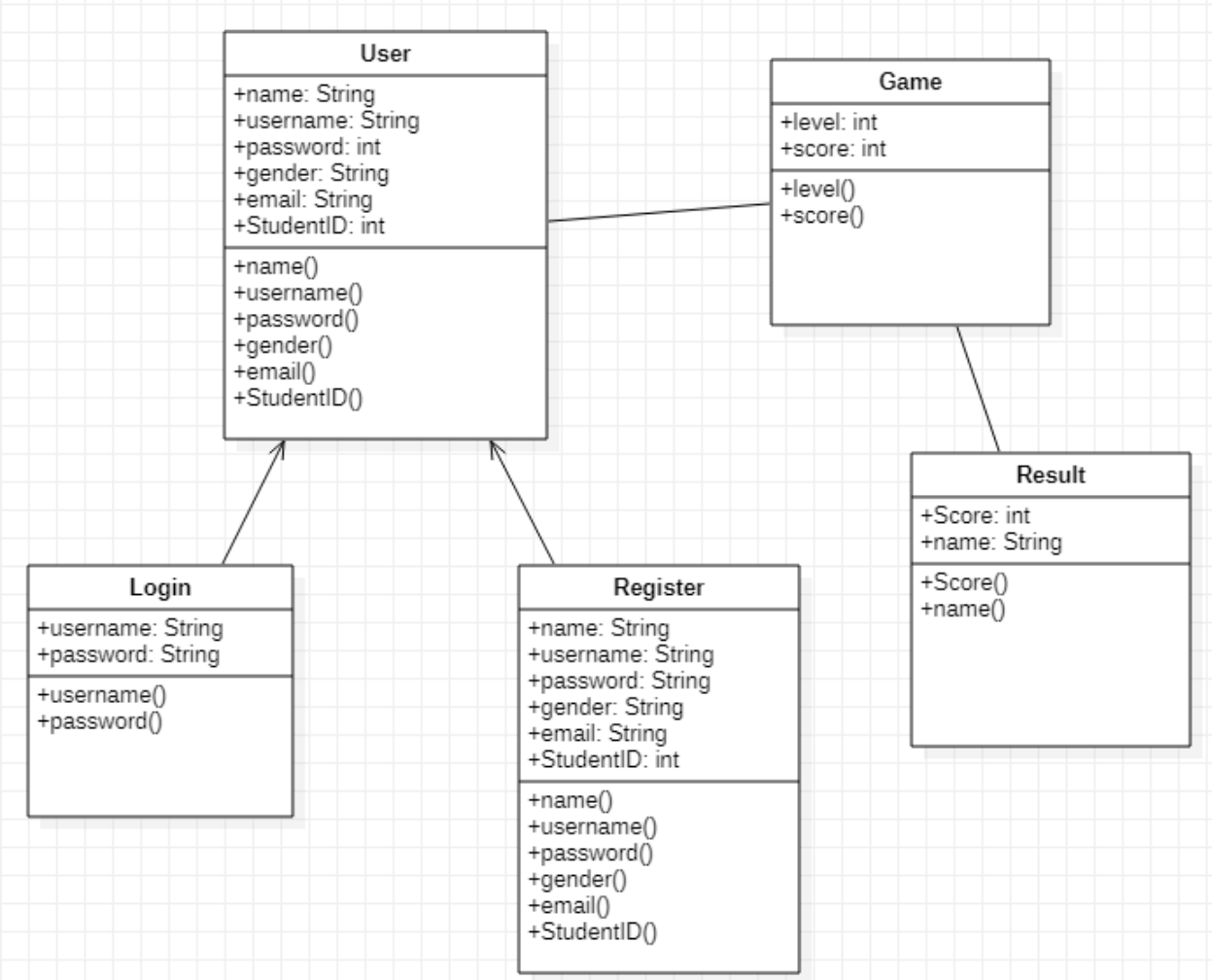


Figure 3.6 Class Diagram

3.3.6 User Interface

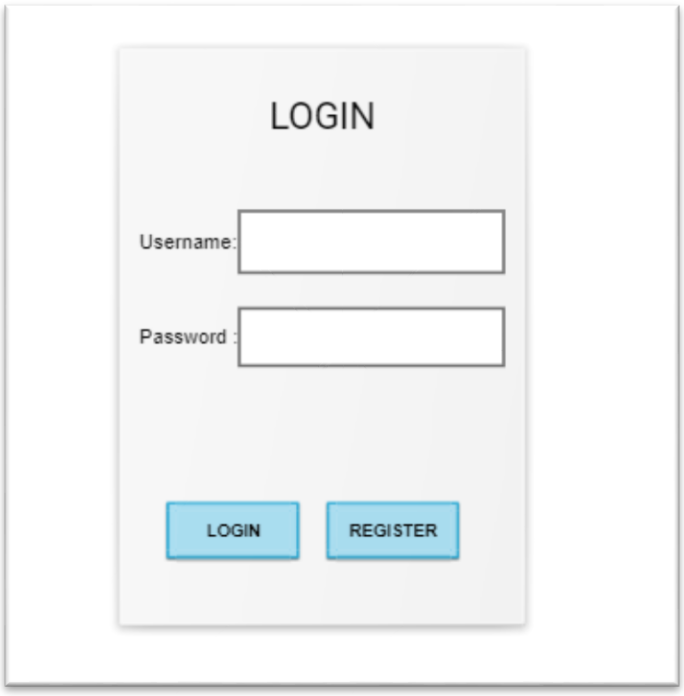


Figure 3.7 Login Page

Figure 3.3.3.1 is the Login page for the user to enter the system. The user has to enter their username, password, and press the login button. But, the new user, they have to register it before login in again. Thus, the new user has to press the register button to register their account.

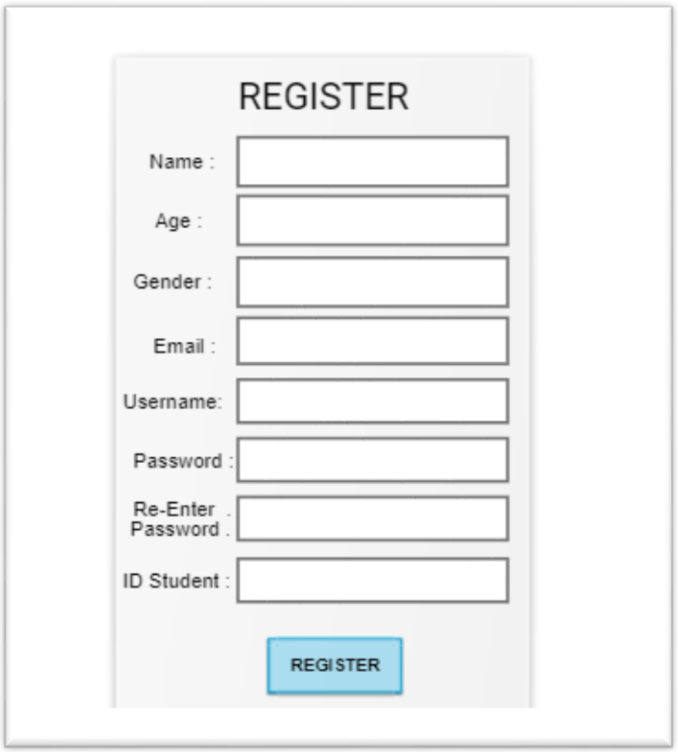


Figure 3.8 Register Page

Figure 3.3.3.2 is the Register Page. This page is created because to register the new user. So, for the new user, they have to fill in the requirements on the registration page which are their name, age, gender, email, create their username, their password, re-enter the password to make sure it and last but not least is the id student.

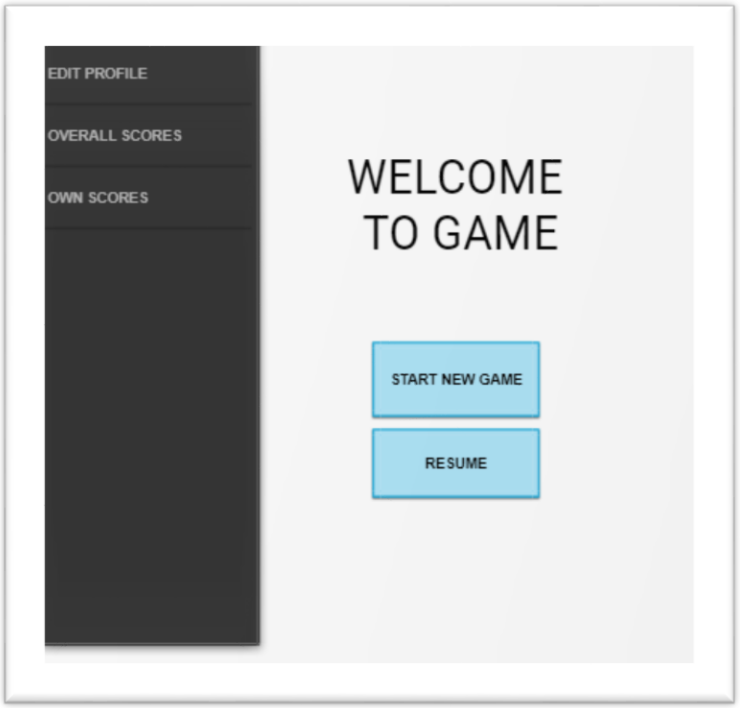


Figure 3.9 Main Page

Figure 3.3.3.3 is the main page. On this page, there have two buttons which are start a new game and resume. If the user wants to play a new game, the user has to press the start new game button but if the user wants to resume the previous game, the user can press the resume button. Other than that, at the left side, there have three buttons which are edit profile, overall scores, and own scores. If the user wants to edit their profile such as their email, they can press the edit profile button. The overall scores button was for the user who wants to see the overall scores that have been collected from the other user and for their own scores is to see their own scores

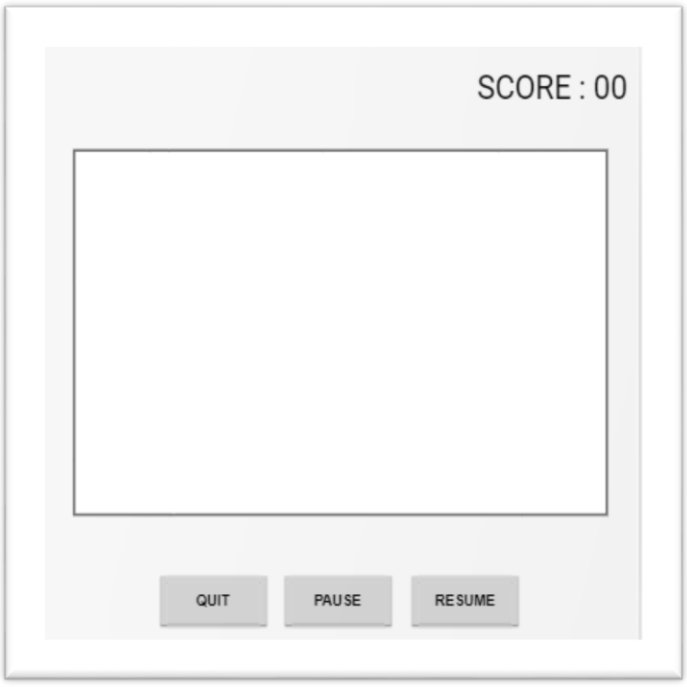


Figure 3.10 Game Page

Figure 3.3.3.4 is the game page. On this page, there have three buttons which quit button, pause button, and resume button. If the user wants to stop playing the game, they can press the quit button. The pause button was for the user that wants to rest or pause the game and for the resume button was when the user wants to continue the game after a pause the game

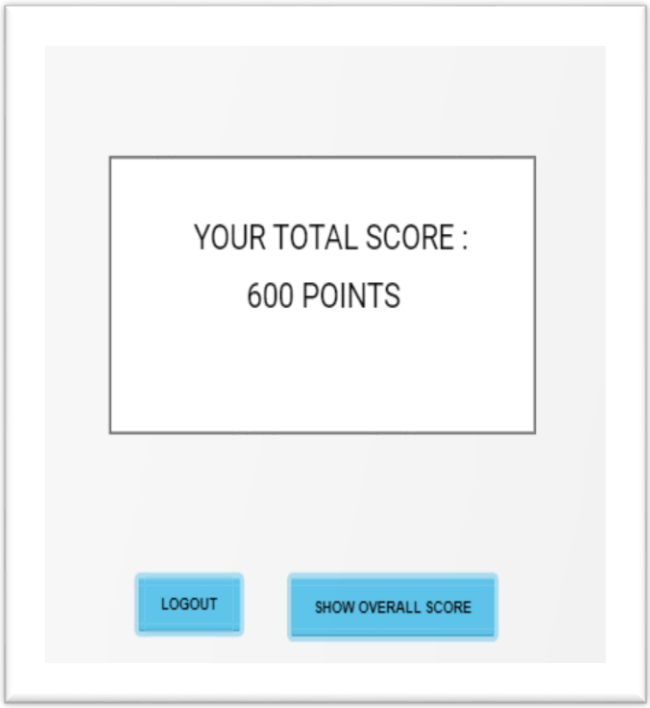


Figure 3.11 Own Score Page

Figure 3.3.3.5 is the own score page. This page would show only the user's score only. In addition, there have two buttons which are the logout button and show overall score button. When the user wants to stop playing a game and want to close the game, he or she has to click the logout button to quit the game. Other than that, show the overall button is for the user who wants to see the overall score that has been collected from the user before this.

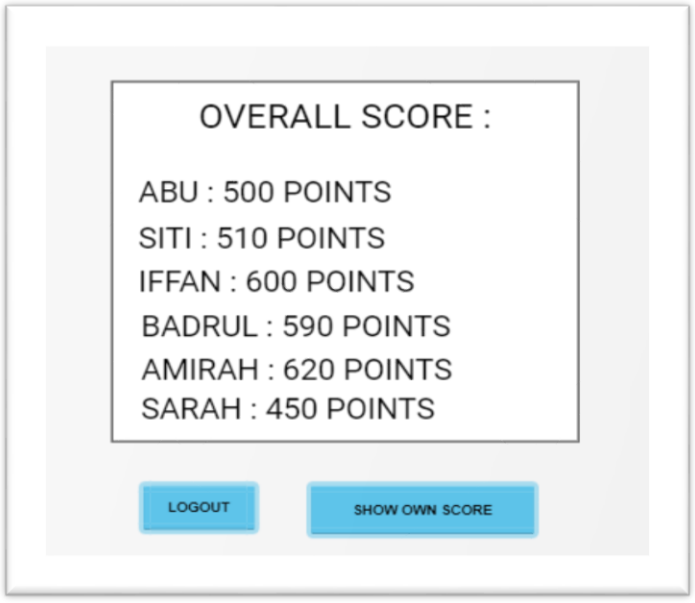


Figure 3.12 Overall Scores Page

Figure 3.3.3.6 is the overall scores page. This page would show the overall scores that have been collected before from different users. Other than that, there have two buttons which are the logout button and show their own score button. The logout button is for the user that wants to quit the game. The show's own score button is for the user that want see their score only

3.4 Development

3.4.1 Development Overview

In this phase, the development of the system will be developed and the function of the system must be well defined. Table 3.1 shows the overall main function of this mobile system.

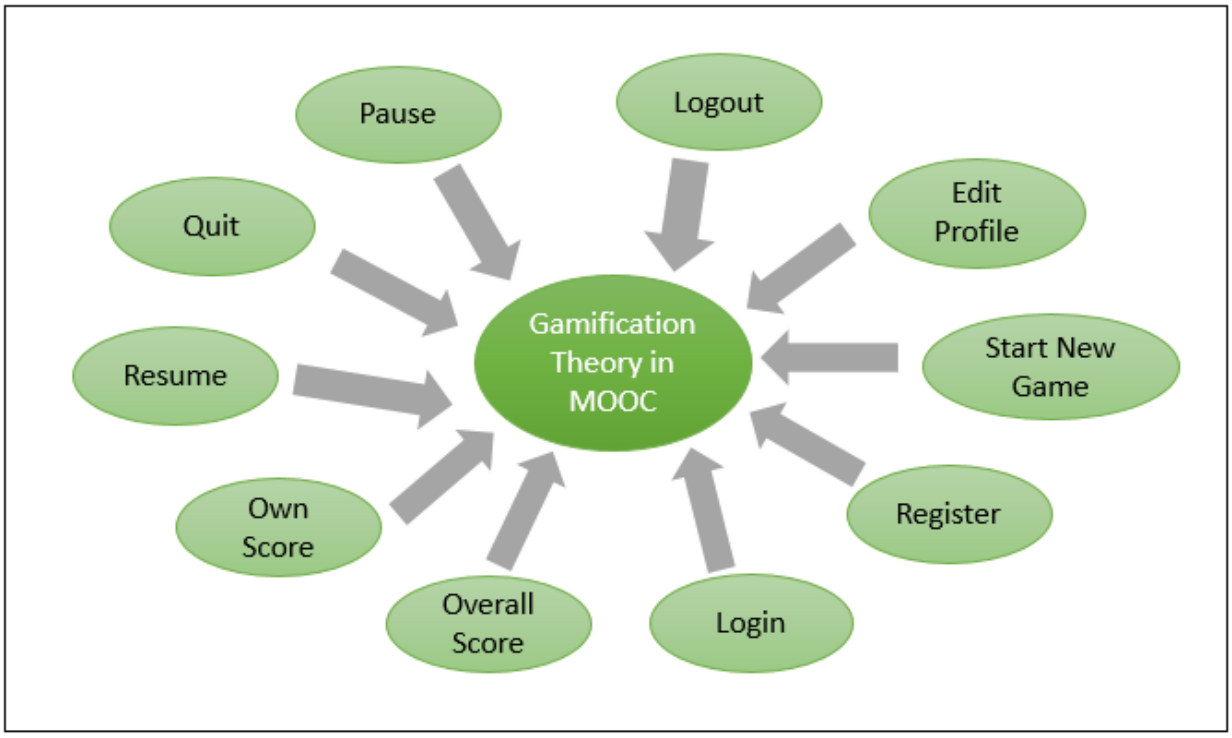


Figure 3.13 Development Overview

Table 3.1 Description of the Development Overview

Function	Description
Start a new game	To start the game on the main page
Login	To use this system, the user has to insert their username and password and click the toggle to enter this system.
Register	It's only for a new user who has to insert their name, age, gender, email, username, password, re-enter the password, and ID student and click the toggle to log in to the system.
Pause	The user able to pause the game when clicking this toggle
Resume	The user can click this toggle when to continue with the previous game or from the game pause
Quit	The user able to quit the game when clicking this toggle
Own Scores	Only show their score when clicking this toggle
Overall Scores	It will show the overall score that has been collected from the other user when clicking this toggle
Edit Profile	To display and edit their profile when clicking this toggle
Logout	The user able to quit the system when clicking this toggle

3.4.2 Modelling for Game Development

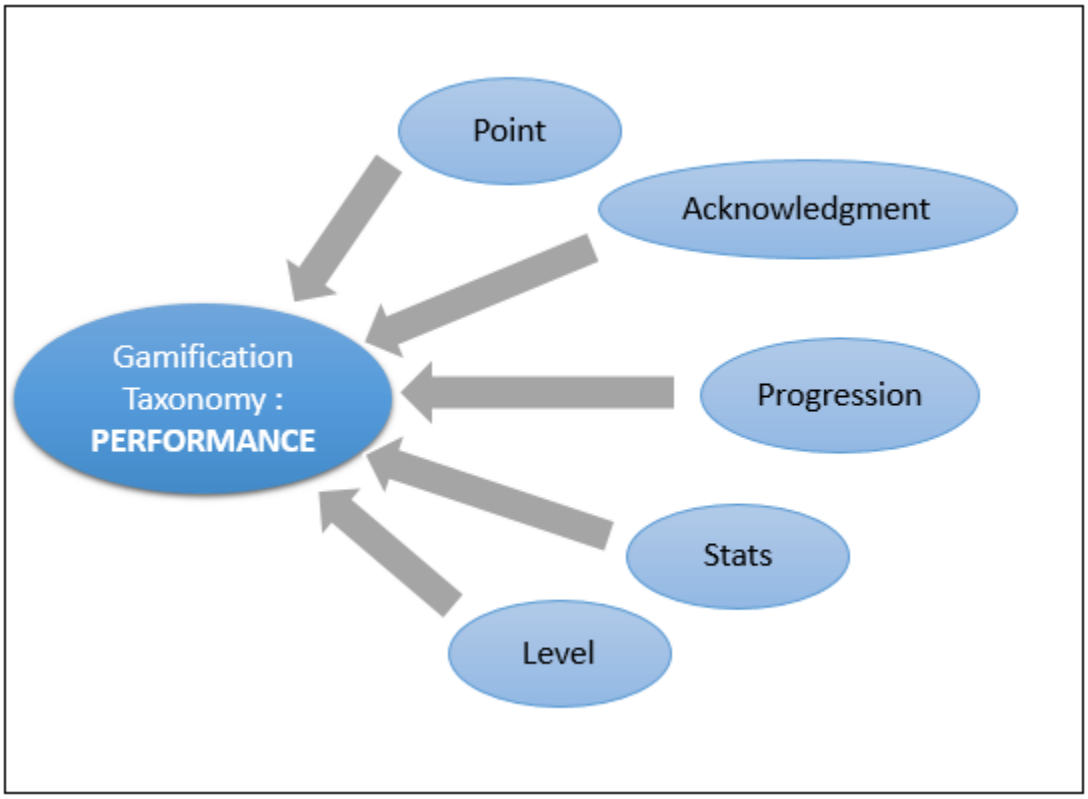


Figure 3.14 Overall Element for The Performance in Gamification Taxonomy

Table 3.2 Description for All Elements in the Performance of Gamification Taxonomy

Elements	Description
Point	It's used to measure the user performs such as scores
Acknowledgment	It's all kinds of feedback that praise the user such as the medal or trophies.
Progression	It allows the user to know their progress within a game such as the progress bar or timer
Stats	Visible information used by the user that related to their outcomes such as their health bar or their life
Level	Hierarchical layer that presents in the game such as the level, skill of level

3.4.3 Hardware and Software Requirements

The hardware and software requirements are to let the user who wants to play know detail about the game before they start the game. They would know specifically about the hardware requirements and software requirements.

Tables 3.1 and 3.2 show the minimum requirement of hardware and software for the developer and the user that is needed in this project.

3.4.3.1 Hardware Requirements

Hardware	Specification for developer	Specification for users
CPU	2.3hz Intel Core i5 processor	2.3hz Intel Core i3 processor
RAM	RAM 8GB DDR4	RAM 4GB DDR2
WINDOWS	Windows 10	Windows 7
TYPE	Gaming Laptop	Ordinary Laptop

Table 3.1 Comparison of hardware requirements between developer and learner.

3.4.3.2 Software Requirements

Software	Specification for developer & users
Unity	Version 2020.2.1

Table 3.2 Software Requirements for the developer and the users

3.5 Testing

After the development phase, the system needs to be tested to verify the system's functionalities and ensure it performs as desired. A survey will be conducted on the user after using the system. it is to observe the user's response to the system. the survey will focus on enjoyment and how long the user would use the system. To make the easy to access an online survey will implement by using Google forms. All the responses from the user will be collected and analyzed so that the system can be improved to make it more enjoyable and the user would engage with the system.

3.6 Maintenance

The maintenance phase is the final phase for the Waterfall Model for game development. The maintenance phase is a challenge that every team developer has to face when this project is deployed to the customer's site. Therefore, the time and effort required to maintain the operating program after launch are very important. Despite the fact, it is a very challenging task, but still the developer has to face it. Also, maintenance phases are a very broad activity that includes error correction, enhancement of capabilities, and optimizations.

3.7 Summary

Phases	Activities	Deliverables
Requirements	• Type of Gamification Taxonomy	• Problem Statement • Objectives • Literature review • Research methodology
Design	• Designing the use case diagram • Designing the flowchart • Designing the User Interface	• Context diagram and process flow diagram
Development	• Software and hardware requirements	• Project
Testing	• Give the testing among the targeted respondents: evaluate their enjoyment and engagement with the system	• Questionnaire • Result
Maintenance	• Correct the error connection to make a quality project	• Maintain the project

Table 3.5 Summary of the Waterfall Model

3.8 Conclusion

Phases that have been discussed according to the Waterfall Model for game development are very important to implement in this project to makes sure this project successful. The goal and objectives can be achieved in this project with the Waterfall Model for game development.

REFERENCES

Alsawaier, R. S. (2018). The effect of gamification on motivation and engagement. *International Journal of Information and Learning Technology*, *35*(1), 56–79. https://doi.org/10.1108/IJILT-02-2017-0009

Dichev, C., & Dicheva, D. (2017). Gamifying education: what is known, what is believed, and what remains uncertain: a critical review. *International Journal of Educational Technology in Higher Education*, *14*(1), 9. https://doi.org/10.1186/s41239-017-0042-5

Gamage, D., Perera, I., & Fernando, S. (2020). MOOCs lack interactivity and collaborativeness: Evaluating MOOC platforms. *International Journal of Engineering Pedagogy*, *10*(2), 94–111. https://doi.org/10.3991/ijep.v10i2.11886

Hanus, M. D., & Fox, J. (2015). Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Computers & Education*, *80*, 152–161. https://doi.org/https://doi.org/10.1016/j.compedu.2014.08.019

Jansen, D., & Schuwer, R. (2015). *Institutional MOOC strategies in Europe* (Issue February).

Klock, A. C. T., Ogawa, A. N., Gasparini, I., & Pimenta, M. S. (2018). Does Gamification Matter? A Systematic Mapping about the Evaluation of Gamification in Educational Environments. *Proceedings of the 33rd Annual ACM Symposium on Applied Computing*, 2006–2012. https://doi.org/10.1145/3167132.3167347

Koivisto, J., & Hamari, J. (2019). The rise of motivational information systems: A review of gamification research. *International Journal of Information Management*, *45*, 191–210. https://doi.org/https://doi.org/10.1016/j.ijinfomgt.2018.10.013

Pappas, C. (2014). The science and the benefits of gamification in elearning. *Retrived from Https://Elearningindustry. Com/Science-Benefits-Gamificationelearning*.

Shi, L., Cristea, A., Hadzidedic, S., & Dervishalidovic, N. (2014). Contextual Gamification of Social Interaction - Towards Increasing Motivation in Social E-learning. *ICWL*.

Szeto, K. Y. (1991). Frustration-mediated superconductivity in two-dimensional classical Heisenberg model. *Bulletin of Materials Science*, *14*(4), 1033–1036. https://doi.org/10.1007/BF02747473

Toda, A. M., do Carmo, R. M. C., da Silva, A. P., Bittencourt, I. I., & Isotani, S. (2019). An approach for planning and deploying gamification concepts with social networks within educational contexts. *International Journal of Information Management*, *46*, 294–303. https://doi.org/https://doi.org/10.1016/j.ijinfomgt.2018.10.001

Toda, A. M., Oliveira, W., Shi, L., Bittencourt, I. I., Isotani, S., & Cristea, A. (2019). Planning gamification strategies based on user characteristics and DM: A gender-based case study. *EDM 2019 - Proceedings of the 12th International Conference on Educational Data Mining*, *i*, 438–443.

Washburne, J. N. (1936). The definition of learning. *Journal of Educational Psychology*, *27*(8), 603.

Palomino, P. T., Toda, A. M., dos Santos, W. O., Cristea, A. I., & Isotani, S. (2019). The narrative for gamification in education: Why should you care? In *Proceedings of the 19th IEEE International Conference on Advanced Learning Technologies.*

Dignan, A. (2011). *Game Frame. Using games as a strategy for success*. Free Press.